



Open ended 3D printers can be augmented to increase their scale of printing by a great degree.

Potential 3: The workflow of traditional manufacturing is based on a sequence of stages where products are created piece by piece and then assembled together. With 3D printing the entire manufacturing workflow is unified into a single step. Since 3D printers can form interlocking elements the restraint of time needed to assemble parts is broken. Even vastly complex assortment of parts such as

those in a gun or a car do not pose a problem for 3D printing and saves significant effort of labour and time which would otherwise be wasted in assembling. This advantage has complementary effects on the supply chain, which helps in saving money otherwise spent on transportation and labour, as well as the fact that they save the environment from pollution.

Potential 4: Unlike traditional printing techniques the amount of time required for planning, and implementation is greatly reduced when working with a 3D printer. Due to this saving we can produce products on demand, as and when it is needed and ready. Even in cases where designs and plans are already prepared, traditional manufacturing has a start up time cost whereas with 3D printing it is just seconds away. Business' and personal manufacturing is immediate and fulfillment is maximised, so far as to say, that even expenses related to the transportation of goods can be minimized since they can be printed closer to the location where they are needed, effectively putting an end to warehousing and storage of inventory.

Potential 5: One of the most practical, and most fun, potential of 3D printing is its openness to the imagination. The requirements of crea-



The Aston Martin used in James Bond's Skyfall was a 3D printed model.